

distinct linear factors

$$\frac{k(s)}{(s + \alpha_1) \cdots (s + \alpha_n)} = \frac{A_1}{s + \alpha_1} + \cdots + \frac{A_n}{s + \alpha_n}$$

repeated linear factors

$$\frac{k(s)}{(s + \alpha)^n} = \frac{A_1}{s + \alpha} + \cdots + \frac{A_n}{(s + \alpha)^n}$$

quadratic factor

$$\frac{k(s)}{(\alpha_1 s^2 + \beta_1 s + \gamma_1)(\alpha_2 s^2 + \beta_2 s + \gamma_2)} = \frac{As + B}{(\alpha_1 s^2 + \beta_1 s + \gamma_1)} + \frac{Cs + D}{(\alpha_2 s^2 + \beta_2 s + \gamma_2)}$$

$$\frac{\omega^2}{1 + (\frac{\omega}{\omega_{p1}})^2} \times \frac{1}{1 + (\frac{\omega}{\omega_{p2}})^2}$$

$$\frac{s}{(s + \omega_{p1})(s + \omega_{p2})} = \frac{A}{s + \omega_{p1}} + \frac{B}{s + \omega_{p2}} = \frac{-\omega_{p1}}{\omega_{p2} - \omega_{p1}} \frac{1}{s + \omega_{p1}} + \frac{\omega_{p2}}{\omega_{p2} - \omega_{p1}} \frac{1}{s + \omega_{p2}}$$

$$s = A(s + \omega_{p2}) + B(s + \omega_{p1})$$

$$1 = A + B \quad 0 = A\omega_{p2} + B\omega_{p1}$$

$$A = 1 - B \quad 0 = (1 - B)\omega_{p2} + B\omega_{p1}$$

$$A = 1 - \frac{\omega_{p2}}{\omega_{p2} - \omega_{p1}} \quad 0 = \omega_{p2} + (-\omega_{p2} + \omega_{p1})B$$

$$= \frac{\omega_{p2} - \omega_{p1} - \omega_{p2}}{\omega_{p2} - \omega_{p1}} \quad B = \frac{\omega_{p2}}{\omega_{p2} - \omega_{p1}}$$

$$= \frac{-\omega_{p1}}{\omega_{p2} - \omega_{p1}}$$

nope

let $x = \omega^2$

$$= \frac{-\frac{\omega_{p1}^2}{\omega_{p2}^2 - \omega_{p1}^2}}{\omega^2 + \omega_{p1}^2} + \frac{\frac{\omega_{p2}^2}{\omega_{p2}^2 - \omega_{p1}^2}}{\omega^2 + \omega_{p2}^2}$$

$$I = \int_0^\infty \dots dt$$

$$= -\frac{1}{\omega_{p2}^2 - \omega_{p1}^2} \int_0^\infty \frac{1}{1 + (\frac{\omega}{\omega_{p1}})^2} dt + \frac{1}{\omega_{p2}^2 - \omega_{p1}^2} \int_0^\infty \frac{1}{1 + (\frac{\omega}{\omega_{p2}})^2} dt$$

$$= -\frac{1}{\omega_{p2}^2 - \omega_{p1}^2} \frac{\omega_{p1}}{2\pi} \times \frac{\pi}{2} + \frac{1}{\omega_{p2}^2 - \omega_{p1}^2} \frac{\omega_{p2}}{2\pi} \times \frac{\pi}{2}$$

$$= \frac{1}{4} \frac{\omega_{p2} - \omega_{p1}}{\omega_{p2}^2 - \omega_{p1}^2}$$

$$= \frac{1}{4} \frac{1}{\omega_{p2} + \omega_{p1}}$$

$$\frac{1}{\omega_{p1}\omega_{p2}} \int_0^\infty \frac{\omega^2}{1 + (\frac{\omega}{\omega_{p1}})^2} \times \frac{1}{1 + (\frac{\omega}{\omega_{p2}})^2} dt = I$$

$$= \frac{1}{4} \frac{\omega_{p1}^2 \omega_{p2}^2}{\omega_{p2} + \omega_{p1}}$$

$$\omega_{p1} = 386k$$

$$\omega_{p2} = 2\pi$$

$$\rightarrow 6.24 \times 10^6$$

$$\omega_{p1} = 386k$$

$$\omega_{p2} = 200k$$

$$\} = 2.54 \times 10^5$$

$$\boxed{\int_0^\infty \frac{\omega^2}{1 + (\frac{\omega}{\omega_{p1}})^2} \times \frac{1}{1 + (\frac{\omega}{\omega_{p2}})^2} dt = \frac{1}{4} \frac{(\omega_{p1} \omega_{p2})^2}{\omega_{p1} + \omega_{p2}}}$$

misc

- don't forget the zero from miller cap
 - must include in gain and phase cal
- see properly, is v_{rms} wanted or v_{rms}^2
- differential noise sources will be doubled when finding input referred
- swing and settling limitations need to check for all cases